Operating Systems

Banana – Unit 4

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Class: 4 K

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1A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

3A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/index
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776320977 6 a.txt
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776321056 6 file1.txt
304444 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776321056 6 file2.txt
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes add file1.txt file2.txt
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes status
Staged changes:
  staged:     a.txt
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  tree.h
  untracked:  pes.c
  untracked:  Makefile
  untracked:  pes.h
  untracked:  commit.h
  untracked:  index.h
  untracked:  test_sequence.sh
  untracked:  tree.c
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  index.c
  untracked:  commit.c
```

3B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/index
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776320977 6 a.txt
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776321056 6 file1.txt
304444 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776321056 6 file2.txt
```

4A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes log
commit ab4886a5049346525e19fe773e0ac9a69c23a374925138425627de724b2f14d4
Author: PES User
Date:   0



commit 8eabb82edad72985d1a4932fbf20cf6b31b5aa764f8e2de316805ab11502f290
Author: PES User
Date:   0


Add farewell


commit 98cb2a69c59d421c3ff74125f26cb72368695f50b1f0f79006db189bd33e1ead
Author: PES User
Date:   0
```

4B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/8e/abb82edad72985d1a4932fbf20cf6b31b5aa764f8e2de316805ab11502f290
.pes/objects/98/cb2a69c59d421c3ff74125f26cb72368695f50b1f0f79006db189bd33e1ead
.pes/objects/ab/4886a5049346525e19fe773e0ac9a69c23a374925138425627de724b2f14d4
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4C.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/refs/heads/main
ab4886a5049346525e19fe773e0ac9a69c23a374925138425627de724b2f14d4
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Full integration test:

```
commit.h  hello.txt  Makefile  pes.c  test_objects  test_tree  tree.h
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 7abee7dfe291... Initial commit

Log after first commit:
commit 7abee7dfe291d0fe8ca66670cf1cb2b2b112f6d73367cc74da45d913d815c278
Author: ARJUN G KANAGAL
Date:   0




--- Second Commit ---
Committed: 93d5536cf8be... Update file.txt

--- Third Commit ---
Committed: 420fb37cc6d2... Add farewell

--- Full History ---
commit 420fb37cc6d22233a4f195565215dfbf706135fefe71cd6d716460cb1ba85e1c
Author: ARJUN G KANAGAL
Date:   0



commit 93d5536cf8befc6d2ab5b3d20998d8c27c98eb25e53750ee04aa6d749169ac43
Author: ARJUN G KANAGAL
Date:   0



commit 7abee7dfe291d0fe8ca66670cf1cb2b2b112f6d73367cc74da45d913d815c278
Author: ARJUN G KANAGAL
Date:   0



--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
420fb37cc6d22233a4f195565215dfbf706135fefe71cd6d716460cb1ba85e1c

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/42/0fb37cc6d22233a4f195565215dfbf706135fefe71cd6d716460cb1ba85e1c
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/7a/bee7dfe291d0fe8ca66670cf1cb2b2b112f6d73367cc74da45d913d815c278
.pes/objects/93/d5536cf8befc6d2ab5b3d20998d8c27c98eb25e53750ee04aa6d749169ac43
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Q5.1 **pes checkout <branch>**

- Read commit hash from .pes/refs/heads/<branch>
- Update .pes/HEAD to:

ref: refs/heads/<branch>

- Load commit → get tree → recreate working directory:
    - Write blobs as files
    - Create directories recursively
    - Remove files not in target tree
- Update index to match tree

Complexity:

- Must safely overwrite working directory
- Handle file changes, deletions, directories
- Prevent data loss (uncommitted changes)

Q5.2 **Detect dirty working directory**

- For each index entry:
    - stat() file → compare mtime and size
    - If different → file modified
- Load target branch tree and compare hashes

Conflict if:

- File modified locally AND
- File exists in target branch AND
- Hash differs

Then abort checkout

Q5.3 **Detached HEAD**

- HEAD points directly to commit, not branch
- New commits are created but not referenced by any branch

Effect:

- Commits become unreachable

Recovery:

pes branch <name> <commit_hash>

Q6.1: Garbage collection

Algorithm:

1. Start from all branch heads
2. Traverse:

commit → tree → blobs

3. Store visited hashes in a **hash set**
4. Delete objects not in set

Estimation:
~100k–300k objects visited (due to shared history)

Q6.2: GC race condition

Problem:

- Commit writes objects but hasn't updated HEAD yet
- GC runs → sees objects as unreachable → deletes them

Result:

- Broken commit (missing objects)

Git avoids this by:

- Atomic writes (write + fsync + rename)
- Running GC separately (not concurrently with commits)